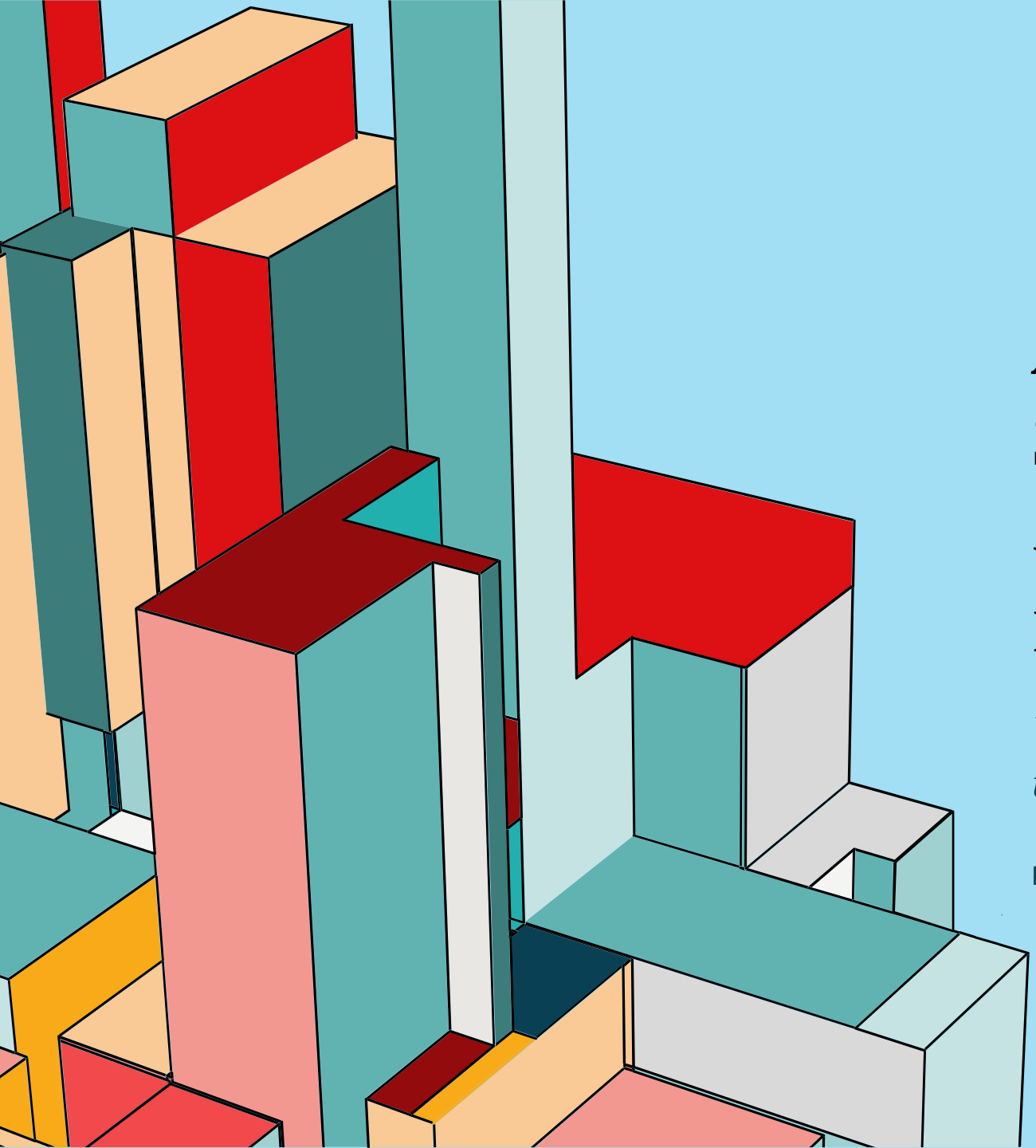


APPLIED DATA SCIENCE PROJECT

UNDERSTANDING VLA FAILURES

When do VLA policies break, and can the training data explain why?

Proposed by **Lorenzo Natale**, Istituto Italiano di Tecnologia (IIT)

Politecnico di Torino

FONDAZIONE links PASSION FOR INNOVATION

ellis European Laboratory for Learning and Intelligent Systems

VALUE-DRIVEN PROJECT

Success rates tell you how often a policy fails, not why

Why

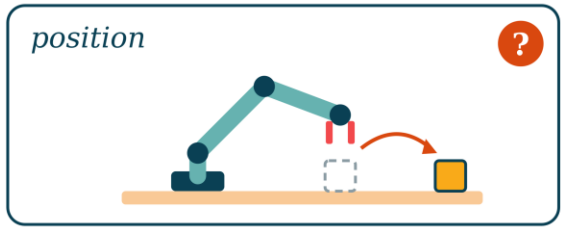
Linking failures to gaps or biases in the training data shows what data to collect next and where a policy can be trusted.

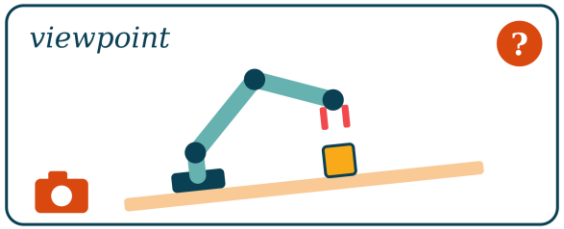
Who benefits

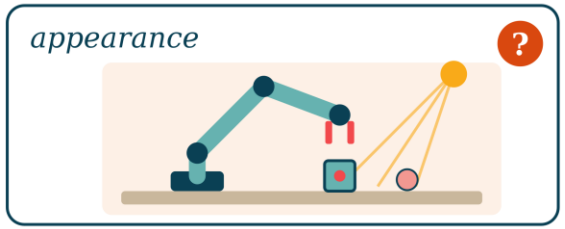
Anyone who relies on robots at work or in daily life, and companies and regulators deciding where robots can be trusted.

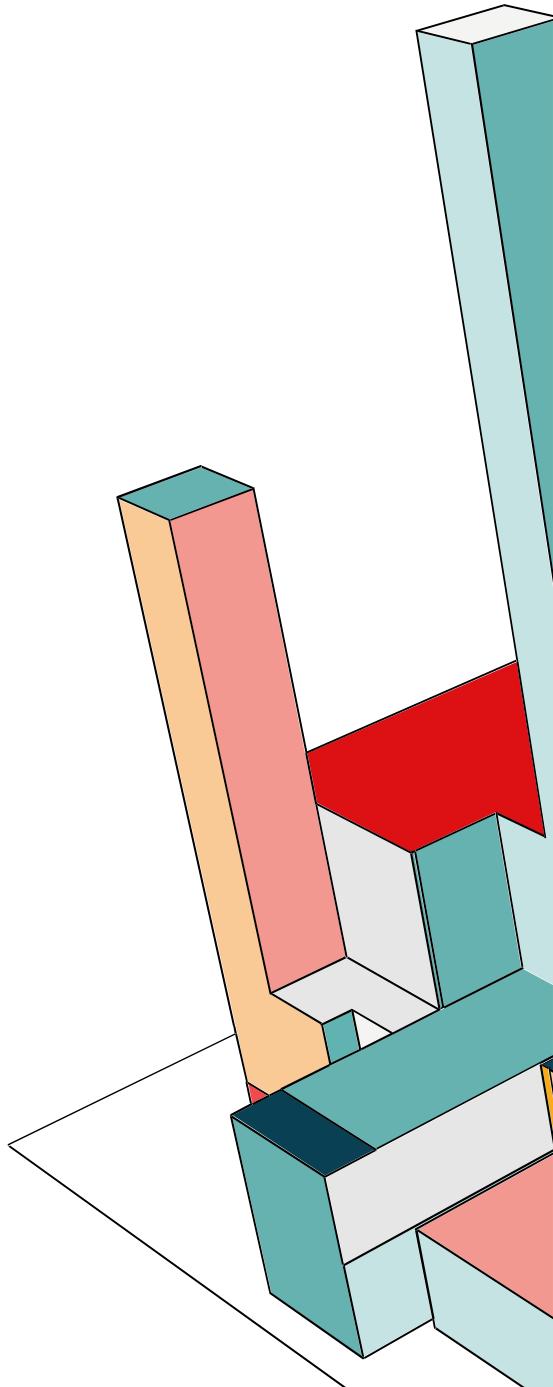
UN SDGs

SDG 9: Industry, Innovation and Infrastructure, through more reliable and transparent robot learning.









DATA

Controlled tests, plus the data the policy learned from

Starting point: a pretrained VLA such as GR00T N1.7, the data it was trained on, and LIBERO-10 tasks or Isaac Sim scenes to vary.

Object positions

Move objects across the workspace, one factor at a time.

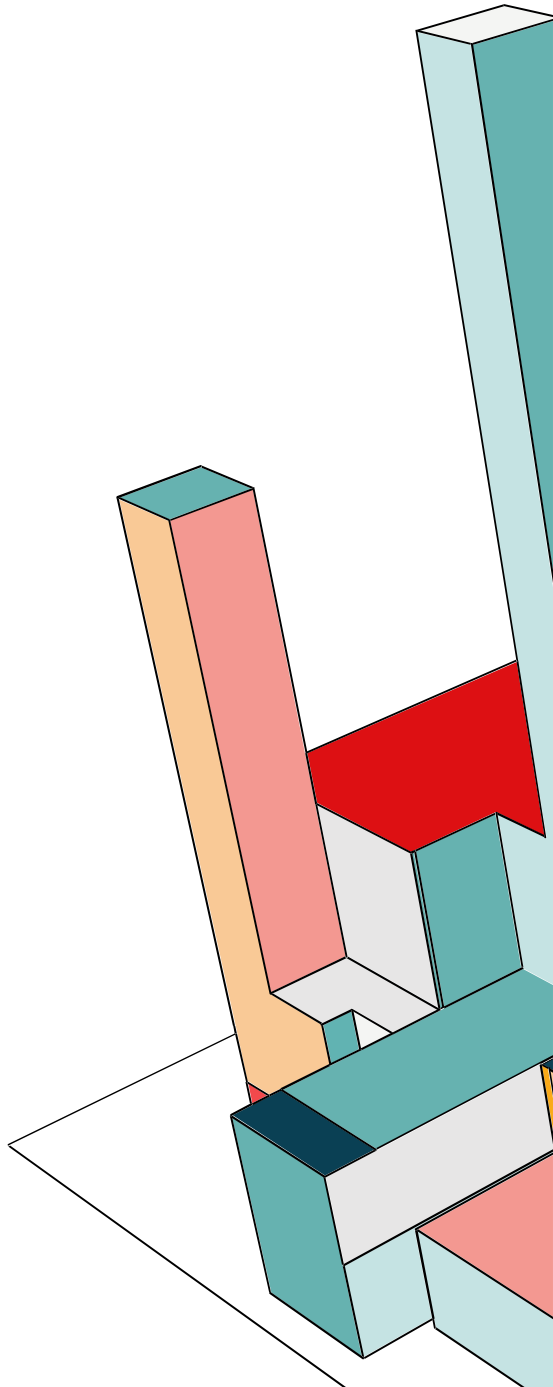
Camera viewpoints

Shift and rotate the camera away from the training setup.

Visual appearance

Change textures, lighting and distractor objects.

Quality: fixed seeds, enough trials for confidence intervals, renders verified, training-data coverage documented.



TASK

Stress-test, diagnose, verify

1

Stress-test

Measure success rate under each controlled variation and pinpoint where the policy fails.

2

Diagnose

Compare failure conditions with the training data's coverage of placements, viewpoints and appearances.

3

Verify

Where resources allow: change the training data, fine-tune, and re-measure to test the explanation.

Metrics: success rate per condition, drop from the nominal setup, training-data coverage of each factor, and change in success after data edits.

